Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Electronics & Telecommunication Engineering

EcoSort

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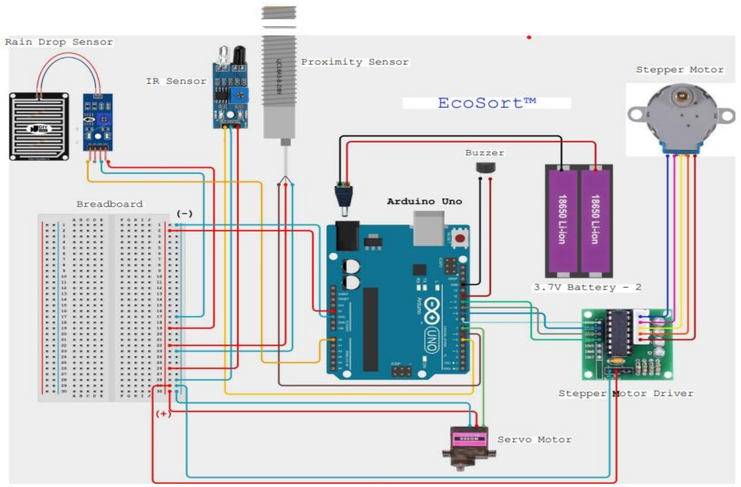
Abstract

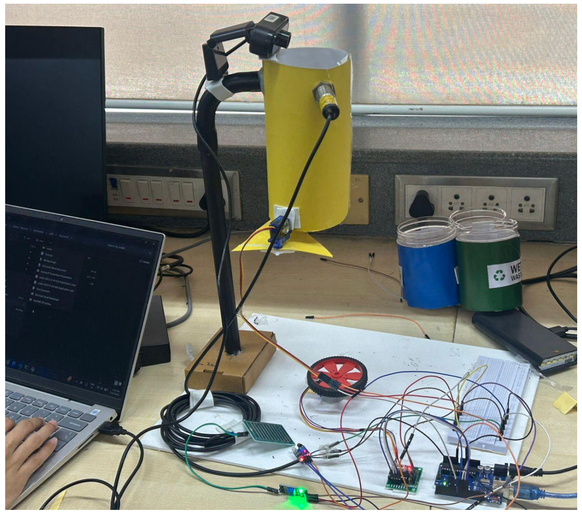
Waste management has become a major environmental challenge due to increasing urbanization and population growth. Improper segregation of waste at the source leads to inefficient recycling and increased landfill usage. Traditional waste segregation methods rely heavily on manual sorting, which is time-consuming and exposes workers to hazardous materials. This paper presents EcoSort, a smart waste management system that automatically classifies waste into metallic, wet, and dry categories using computer vision and machine learning techniques.

Proposed Solution

EcoSort system proposes a computer vision-based waste classification approach. The system uses a webcam to capture images of the waste item placed in the bin. These images are processed using a YOLOv11 object detection model, which identifies the type of waste. Once the waste category is detected, the system activates a motor-driven sorting mechanism that directs the waste into the appropriate compartment for metallic, wet, or dry waste. This approach enables intelligent and automated waste segregation using real-time image processing and machine learning techniques.

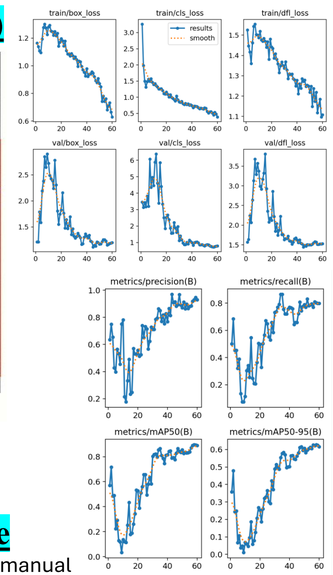
Block Diagram: System Architecture / Circuit Diagram





Simulation (Object Detection)





Conclusions and Future Scope

- The proposed system reduces the need for manual waste segregation and improves the efficiency of recycling processes.
- The integration of machine learning techniques allows the system to identify waste based on visual features, providing greater flexibility compared to sensor-based methods.
- Future improvements may include expanding the dataset to improve detection accuracy, integrating IoT connectivity for monitoring waste generation patterns, and optimizing the system for large-scale deployment in smart city environments.

References

[1] A. B. Wahyutama and M. Hwang, "YOLO-Based Object Detection for Separate Collection of Recyclables and Capacity Monitoring of Trash Bins," *Electronics*, vol. 11, no. 9, p. 1323, 2022, doi: 10.3390/electronics11091323.

[2] M. H. Moktar, H. Mohamed, S. S. H. Hajjaj, and M. Z. Baharuddin, "Medical Waste Sorting Machine Development with IoT and YOLO Model Utilization," *Journal of Engineering and Applied Science*, vol. 72, no. 1, 2025, doi: 10.1186/s44147-025-00661-5.